Masih Haseli

CURRICULUM VITAE

Department of Computing and Mathematical Sciences
California Institute of Technology

mhaseli@caltech.edu
mhaseli.github.io

RESEARCH

- Dynamical Systems and Nonlinear Control
- Koopman Operator Theory and Data-Driven Modeling
- Machine Learning
- Robotics and Robot Learning

EMPLOYMENT

Postdoctoral Scholar Research Associate — Jul. 2025 – present
Department of Computing and Mathematical Sciences
California Institute of Technology
Advisor: Prof. Joel W. Burdick

Postdoctoral Scholar — Sep. 2022 – Jun. 2025
Department of Mechanical and Aerospace Engineering
University of California, San Diego
Advisor: Prof. Jorge Cortés

EDUCATION

Ph.D. in Engineering Sciences (Mechanical Engineering) — Sep. 2017 – Aug. 2022
University of California, San Diego
Advisor: Prof. Jorge Cortés

M.Sc. in Electrical Engineering – Control — Sep. 2013 – Oct. 2015
Amirkabir University of Technology, Tehran

B.Sc. in Electrical Engineering – Control — Sep. 2009 – Sep. 2013
Amirkabir University of Technology, Tehran

HONORS & AWARDS

IEEE Control Systems Letters Outstanding Paper Award — 2025
IEEE Control Systems Society

Robert Skelton Systems and Control Best Ph.D. Dissertation Award — 2023
UCSD Center for Control Systems and Dynamics

Best Student Paper Award — 2021
The 2021 American Control Conference, New Orleans, Louisiana

Bronze Medal — 2014
Iran's National Mathematics Competition

Silver Medal — 2008
Iran's National Physics Olympiad

PUBLICATIONS

Journal Articles and Manuscripts

J1 **Modeling nonlinear control systems via Koopman control family: universal forms and subspace invariance proximity**
M. Haseli, J. Cortés
Automatica 185 (2026), 112722

J2 **On the existence of Koopman linear embeddings for controlled nonlinear systems**
X. Shang, **M. Haseli**, J. Cortés, Y. Zheng
IEEE Transactions on Automatic Control, *submitted*

J3 **Two roads to Koopman operator theory for control: infinite input sequences and operator families**
M. Haseli, I. Mezić, J. Cortés
IEEE Transactions on Automatic Control, *submitted*

J4 **Koopman operators in robot learning**
L. Shi, **M. Haseli**, G. Mamakoukas, D. Bruder, I. Abraham, T. Murphey, J. Cortés, K. Karydis
IEEE Transactions on Robotics 42 (2026), 1088–1107

J5 **Recursive forward-backward EDMD: guaranteed algebraic search for Koopman invariant subspaces**
M. Haseli, J. Cortés
IEEE Access 13 (2025), 61006–61025

J6 **Invariance proximity: closed-form error bounds for finite-dimensional Koopman-based models**
M. Haseli, J. Cortés
Systems & Control Letters, *submitted*

J7 **Generalizing dynamic mode decomposition: balancing accuracy and expressiveness in Koopman approximations**
M. Haseli, J. Cortés
Automatica 153 (2023), 111001

J8 **Temporal forward-backward consistency, not residual error, measures the prediction accuracy of extended dynamic mode decomposition**
M. Haseli, J. Cortés
IEEE Control Systems Letters 7 (2023), 649–654
IEEE Control Systems Letters Outstanding Paper Award Winner

J9 **Learning Koopman eigenfunctions and invariant subspaces from data: Symmetric Subspace Decomposition**
M. Haseli, J. Cortés
IEEE Transactions on Automatic Control 67 (7) (2022), 3442–3457

J10 **Parallel learning of Koopman eigenfunctions and invariant subspaces for accurate long-term prediction**
M. Haseli, J. Cortés
IEEE Transactions on Control of Network Systems 8 (4) (2021), 1833–1845

Conference Proceedings

C1 **Control forward-backward consistency: quantifying the accuracy of Koopman control family models**
M. Haseli, J. Cortés, J. W. Burdick
Proceedings of the IEEE Conference on Decision and Control, Honolulu, Hawaii, 2026, *to appear*

C2 **Koopman operator extensions for control: bridging infinite input sequences and operator families**
M. Haseli, I. Mezić, J. Cortés
Proceedings of the 64th IEEE Conference on Decision and Control, Rio de Janeiro, Brazil, 2025, pp. 1741–1746

C3 **Real-time learning of predictive dynamic obstacle models for robotic motion planning**
S. Kombo, **M. Haseli**, S. X. Wei, J. W. Burdick
Proceedings of the IEEE International Conference on Robotics and Automation, Vienna, Austria, 2026, *to appear*

C4 **Data-driven approximation of Koopman-invariant subspaces with tunable accuracy**
M. Haseli, J. Cortés
Proceedings of the American Control Conference, New Orleans, Louisiana, 2021, pp. 470–475
Best Student Paper Award Winner

C5 **Fast identification of Koopman-invariant subspaces: parallel symmetric subspace decomposition**
M. Haseli, J. Cortés
Proceedings of the American Control Conference, Denver, Colorado, 2020, pp. 4545–4550

C6 **Efficient identification of linear evolutions in nonlinear vector fields: Koopman invariant subspaces**
M. Haseli, J. Cortés
Proceedings of the IEEE Conference on Decision and Control, Nice, France, 2019, pp. 1746–1751

C7 **Approximating the Koopman operator using noisy data: noise-resilient extended dynamic mode decomposition**
M. Haseli, J. Cortés
Proceedings of the American Control Conference, Philadelphia, PA, 2019, pp. 5499–5504

TEACHING EXPERIENCE

Nonlinear Control (UCSD MAE 281B) — Spring 2021
Graduate Teaching Assistant
Instructor: Prof. Jorge Cortés

INVITED TALKS

Title: The Koopman Operator: Truths, Half-Truths, and Beautifully Packaged Lies — Jun. 2026
Invited Seminar
Department of Mechanical and Aerospace Engineering, University of California, Irvine

Title: Koopman Control Family and Universal Finite-Dimensional Forms — May 2025
SIAM Conference on Applications of Dynamical Systems, Denver, Colorado

Title: Closed-Form Error Bounds for Finite-Dimensional Koopman-Based Models and Implications for Learning — Mar. 2024
U.S. Association for Computational Mechanics, Student Chapter Seminars
Online (YouTube)

Title: Koopman Control Family and Universal Finite-Dimensional Forms — Jan. 2024
Safe Autonomous Systems Lab Seminars
Department of Mechanical and Aerospace Engineering, University of California, San Diego

Title: Learning Koopman Eigenfunctions and Invariant Subspaces from Data — Sep. 2023
Scalable Optimization and Control Lab Seminars
Department of Electrical and Computer Engineering, University of California, San Diego

Title: Learning Koopman Eigenfunctions and Invariant Subspaces from Data: Symmetric Subspace Decomposition — Dec. 2022
2022 International Symposium on Nonlinear Theory and Its Applications

Title: Tuning Accuracy in Data-Driven Learning of Unknown Dynamical Systems via the Koopman Operator — Sep. 2021
Mechanistic Machine Learning and Digital Twins for Computational Science, Engineering & Technology Conference

Title: Efficient Identification of Linear Evolutions in Nonlinear Vector Fields: Koopman Invariant Subspaces — Jan. 2020
37th Southern California Control Workshop, University of California, San Diego

PROFESSIONAL SERVICE

Reviewer for: Automatica, IEEE Open Journal of Control Systems, IEEE Access, IEEE Control Systems Letters, Physica D: Nonlinear Phenomena, Journal of Dynamic Systems, Measurement, and Control, IEEE Conference on Decision and Control (CDC), American Control Conference (ACC), International Symposium on Mathematical Theory of Networks and Systems (MTNS), Resilience Week Symposium, Indian Control Conference, IFAC World Congress, International Conference on Robotics and Automation (ICRA), The Journal of Supercomputing, Mathematics of Control, Signals, and Systems